Causal Inference

Bayesian Statistics 2025/26

Aleksij Kraljič

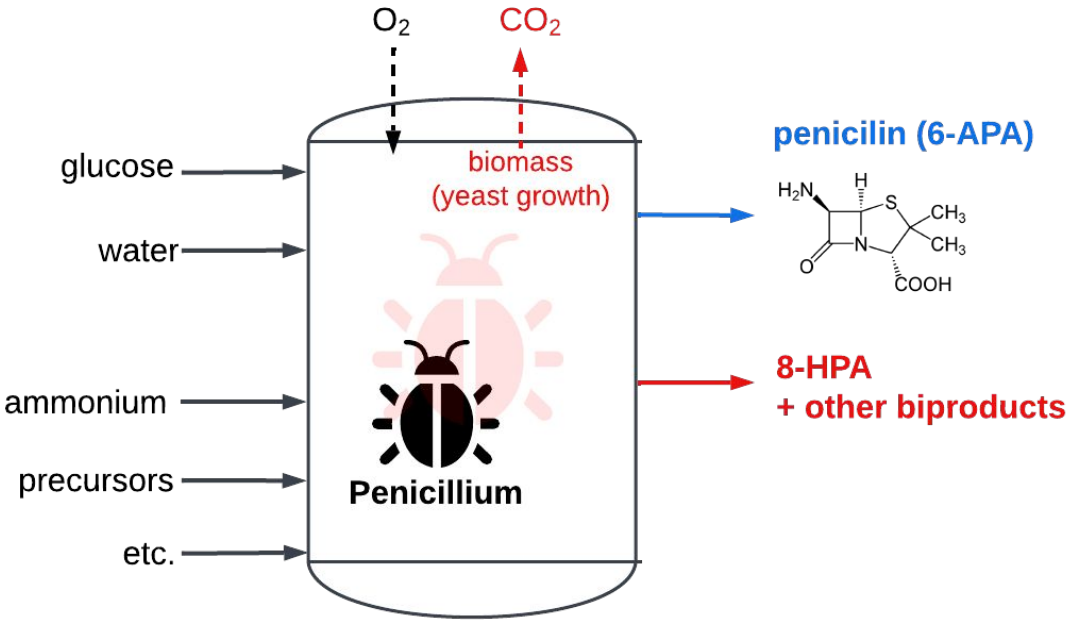
January 6, 2026

Introduction

- Causal inference seeks to move beyond correlation to establish cause and effect.
- **Causal relationships derived from observational data are entirely assumed.**
- We use a case study to explain variability in total penicillin output.
- The goal is to derive meaningful regression models **(pick explanatory variables wisely)** to gain some insights in the outcome variability.

Part I: Case Study

Background: penicillin production

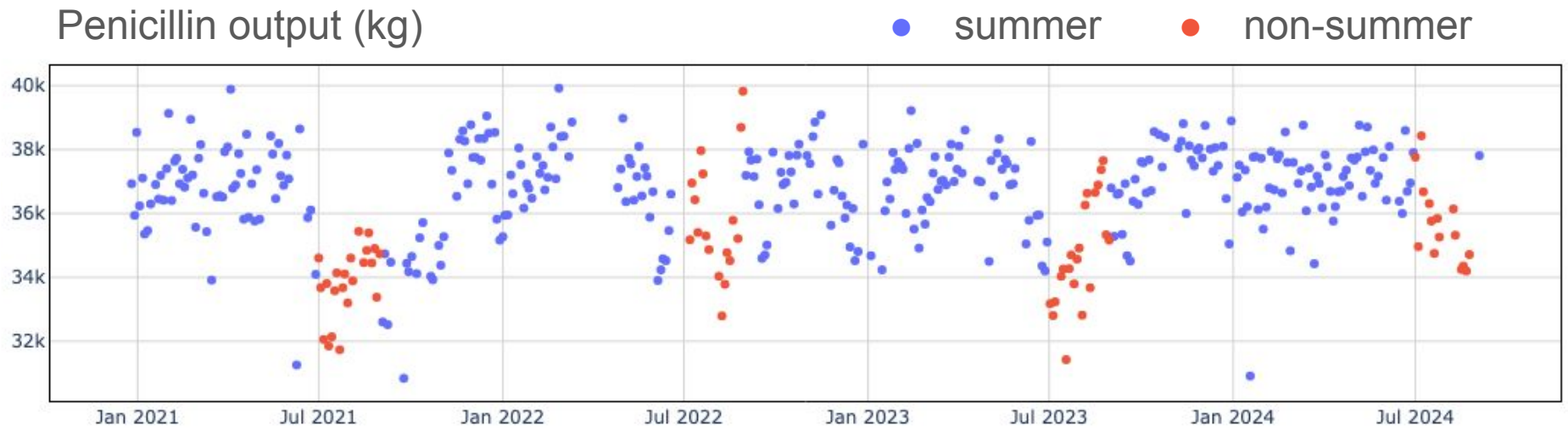


Key performance indicators

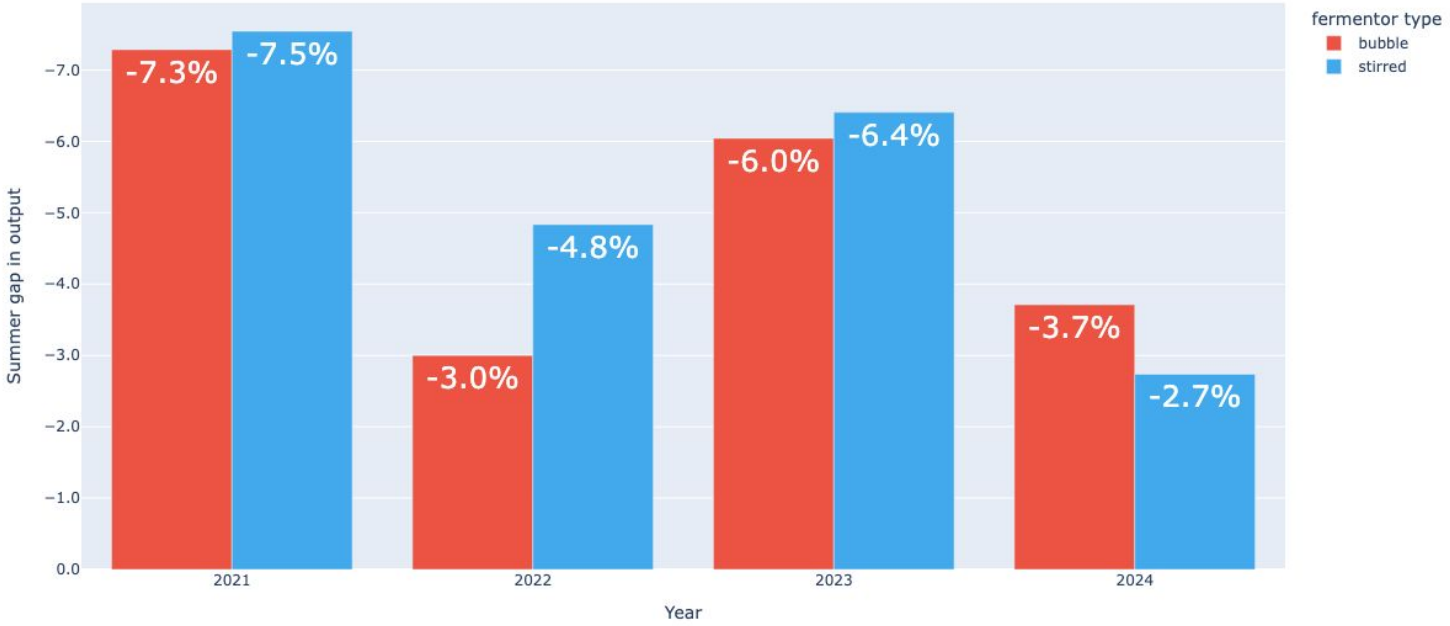
- output
  - the amount of penicillin
- yield (efficiency)
  - output per added sugar

The problem → output drop in summer months

The goal → explain this variation

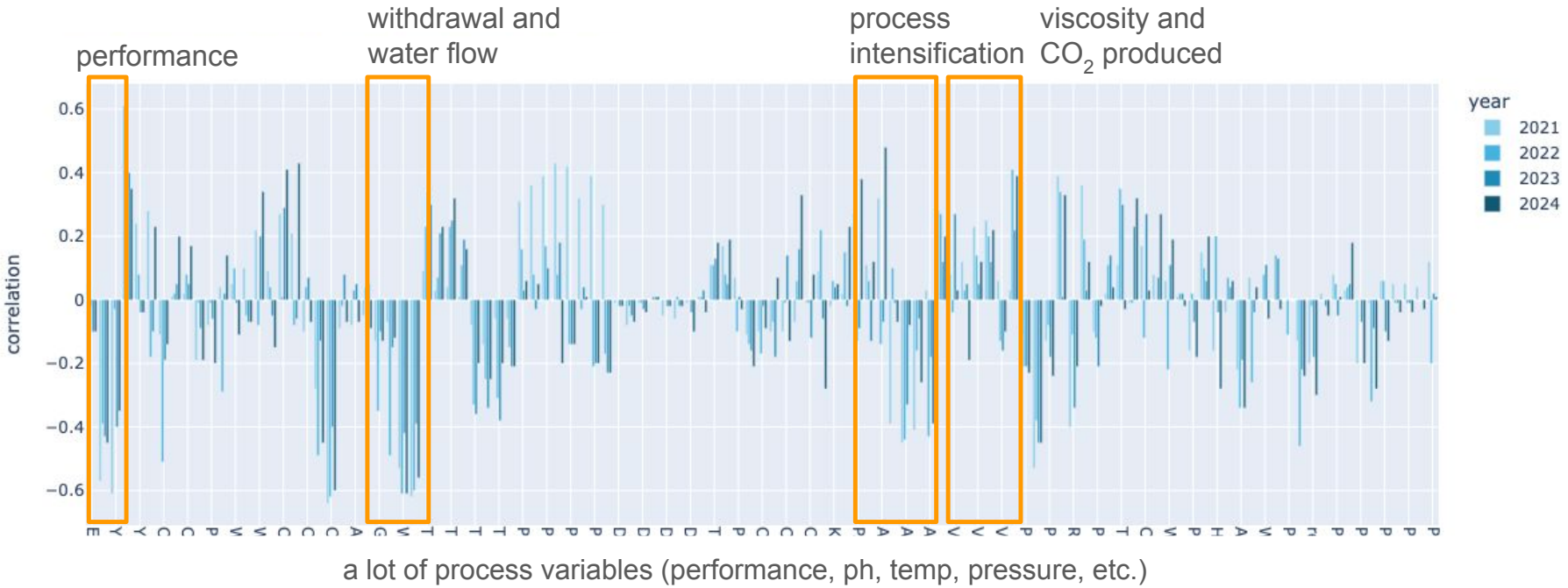


Why? Because 1% of a batch costs $1,000,000!




```
gap in % = (summer-non_summer) / non_summer *100%
```

Exploratory analysis - correlation profiles

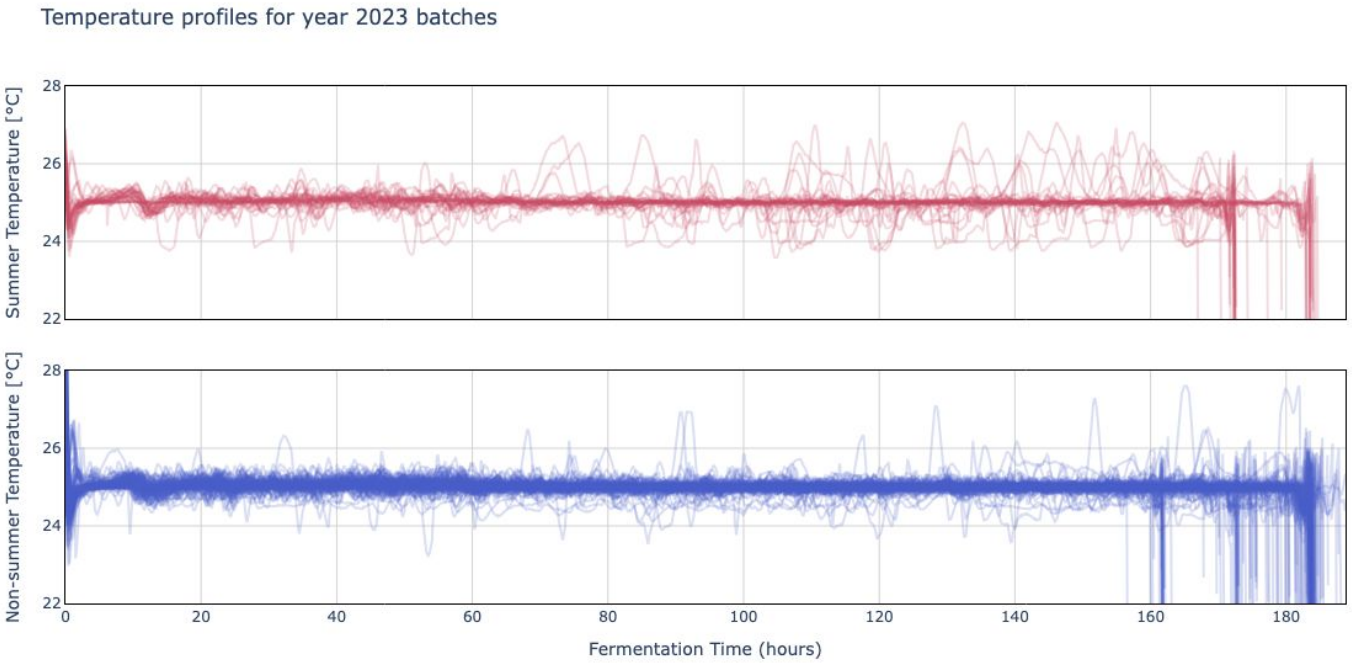


Exploratory analysis



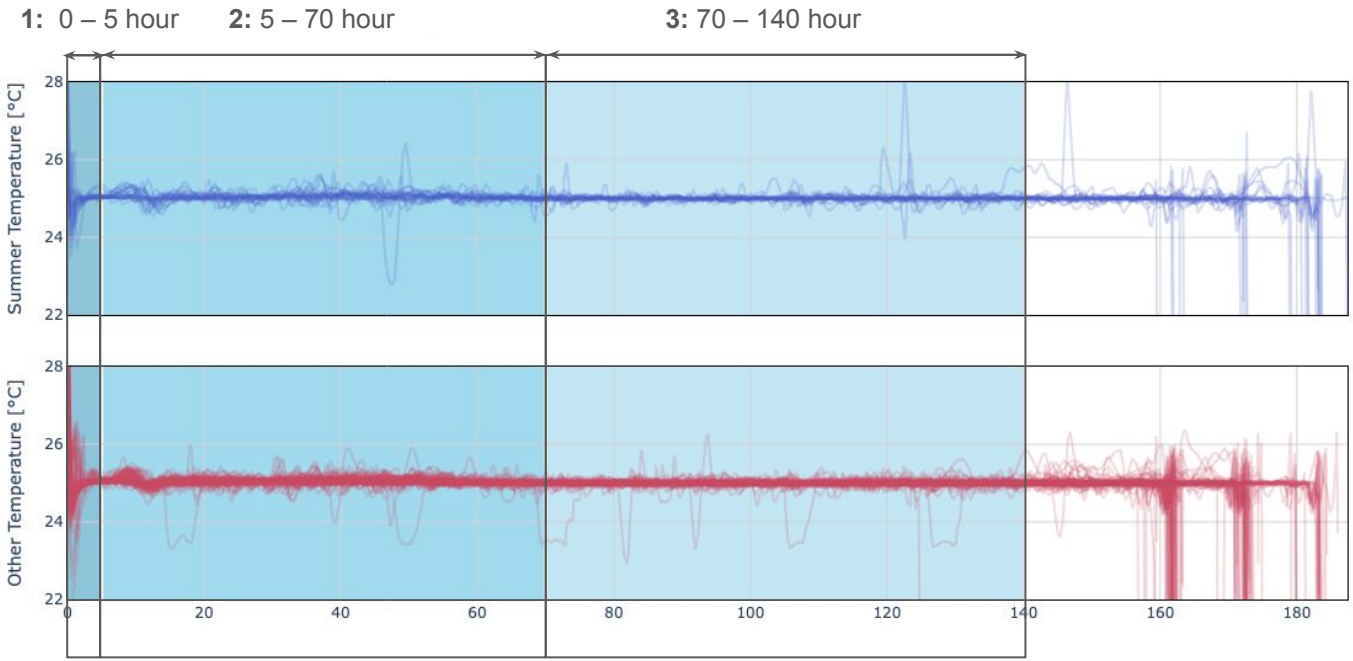
A clear negative correlation with key performance indicators.

Exploratory analysis



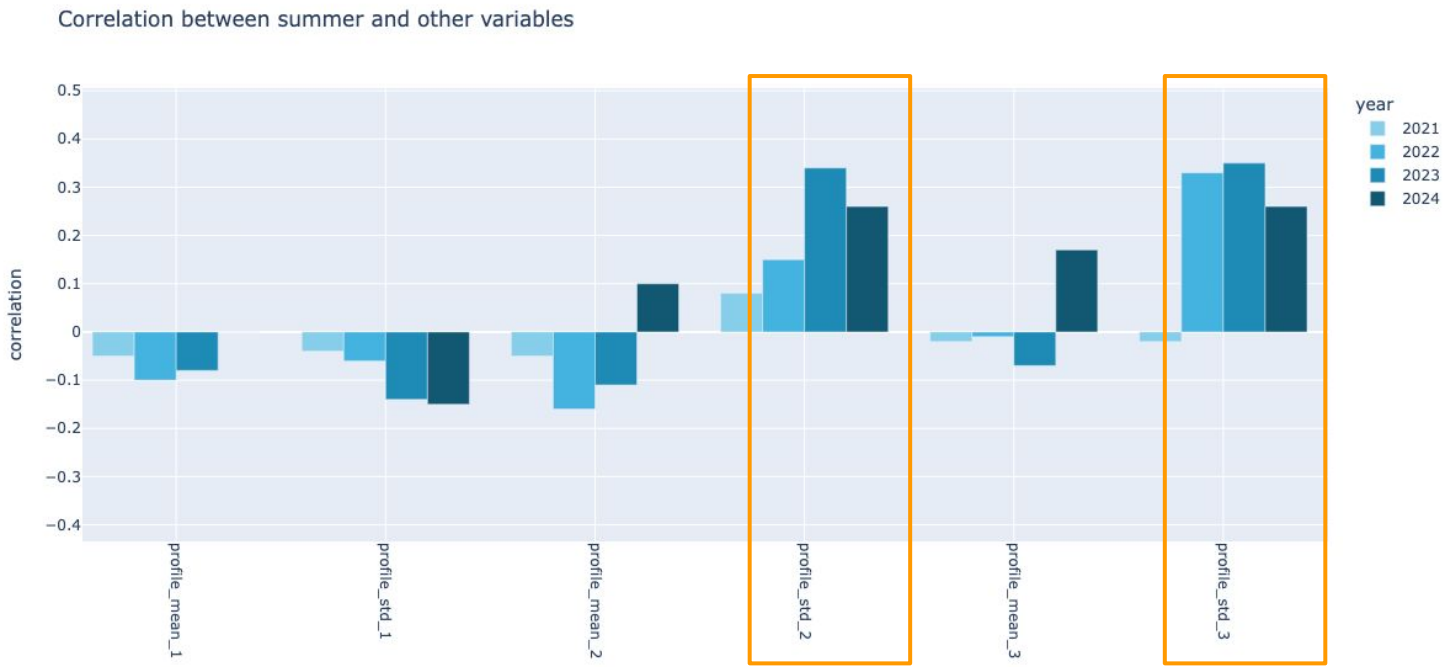
Higher temperature variation observed during summer months.

Broth temperature feature extraction



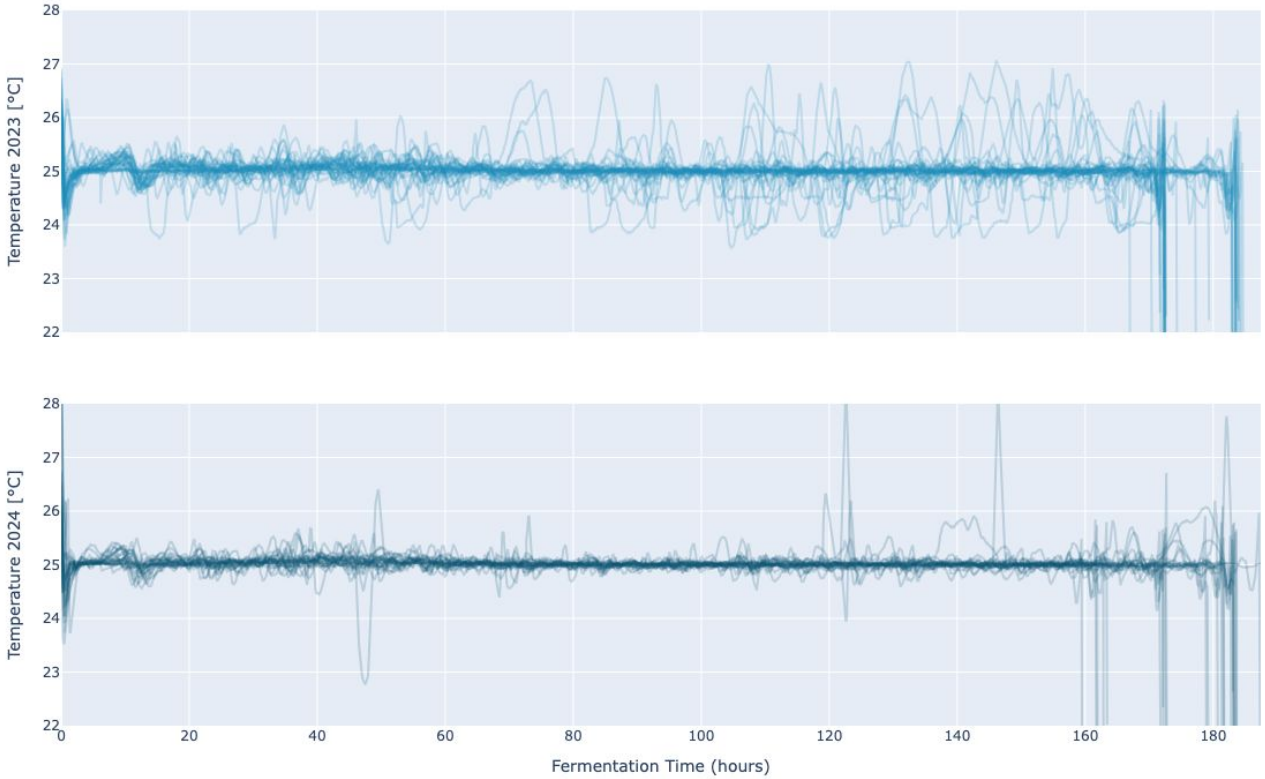
Temperature profiles time window definition.

Temperature correlations with summer months



Temperature profile standard deviation exhibits positive correlations with summer months.

Summer temperature profiles comparison



Large difference between the summer batches of 2023 and 2024.

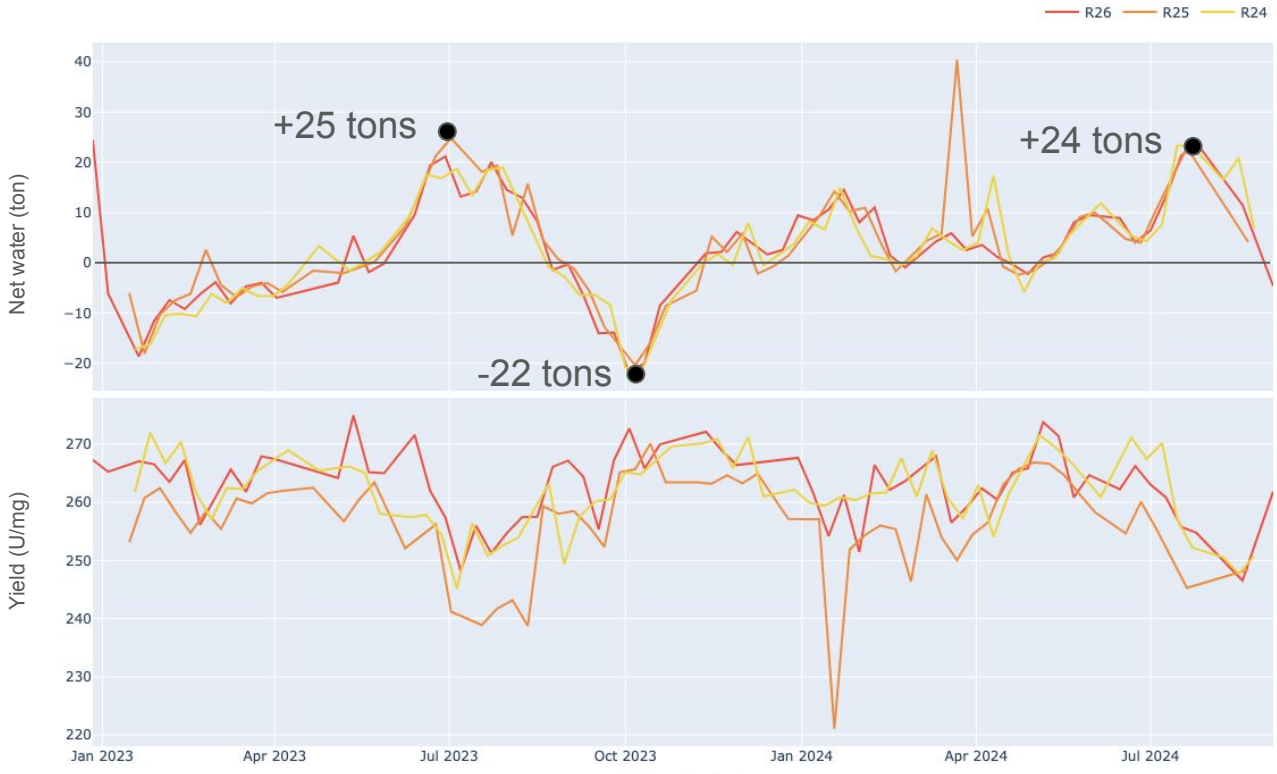
Relationship between humidity and added water



A negative relationship between fermentation air water content and total water added is evident, although there are some deviations.

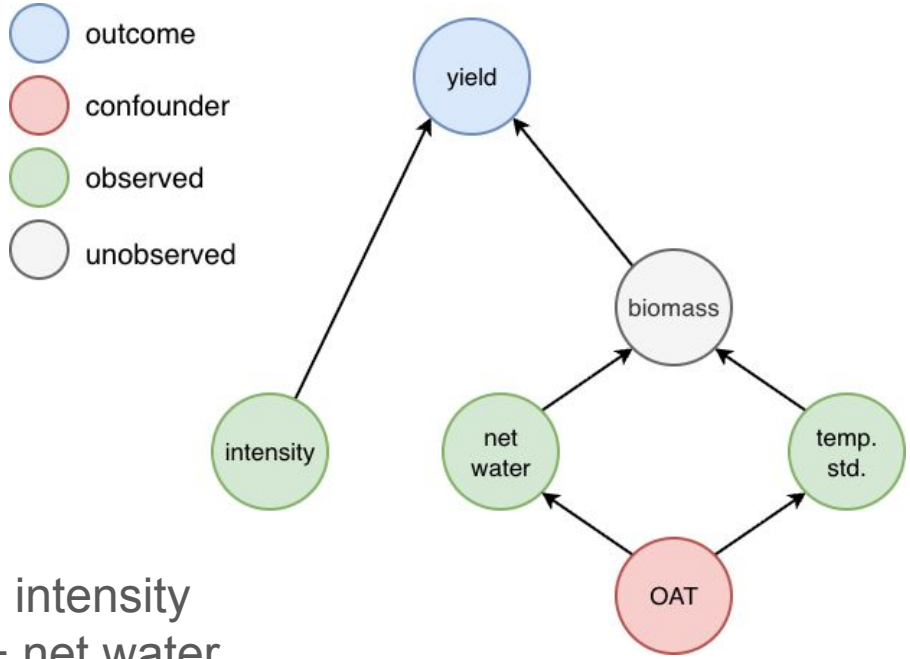
Expected drop at 10 g/kg → What is the reason behind other variations?

Net water spans a wide range between seasons



In summer, fermentors receive up to 25 tons of additional water, despite very low manually added water, while in winter net water reaches -22 tons.

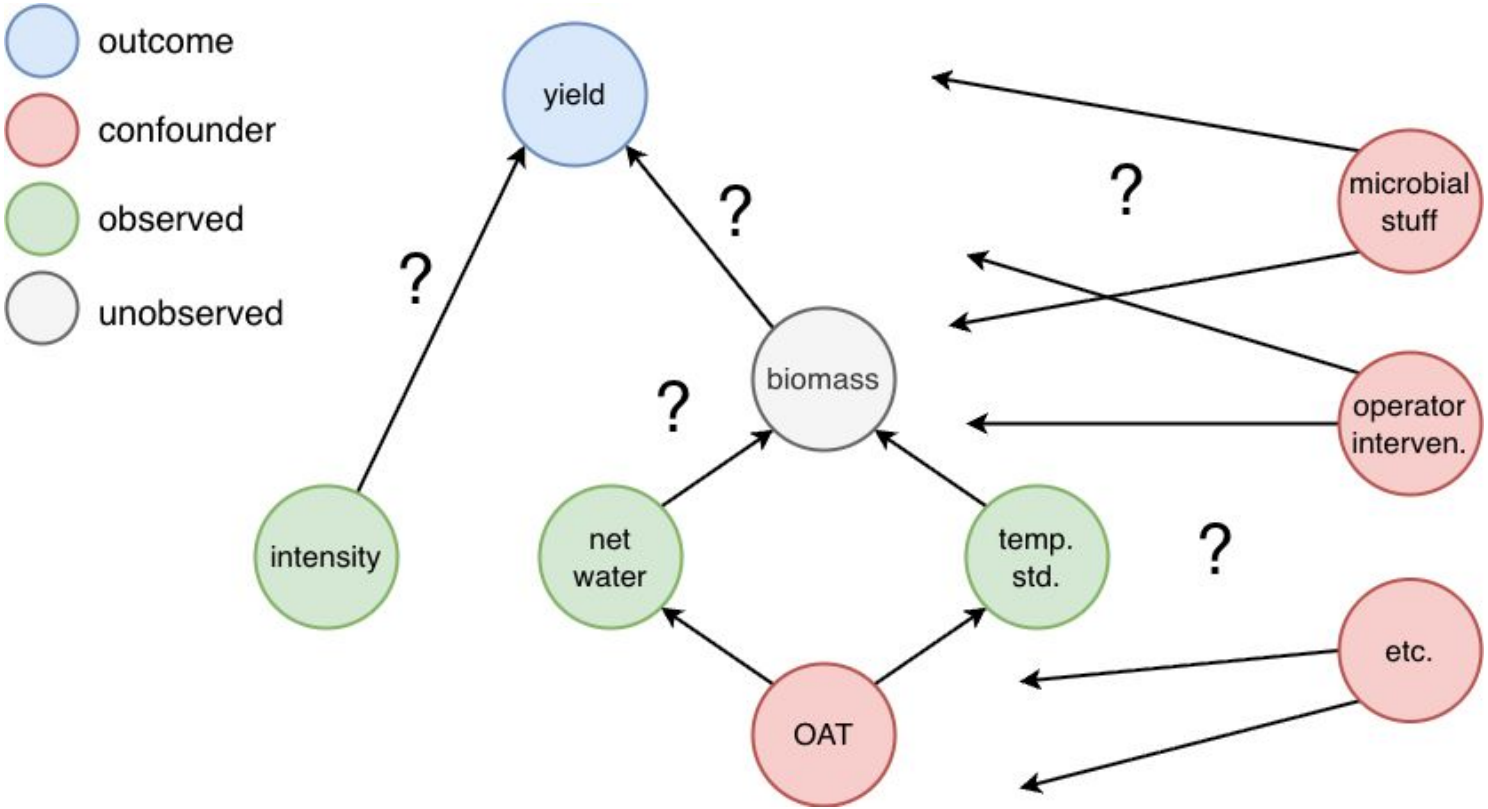
DAG - the assumed causal model



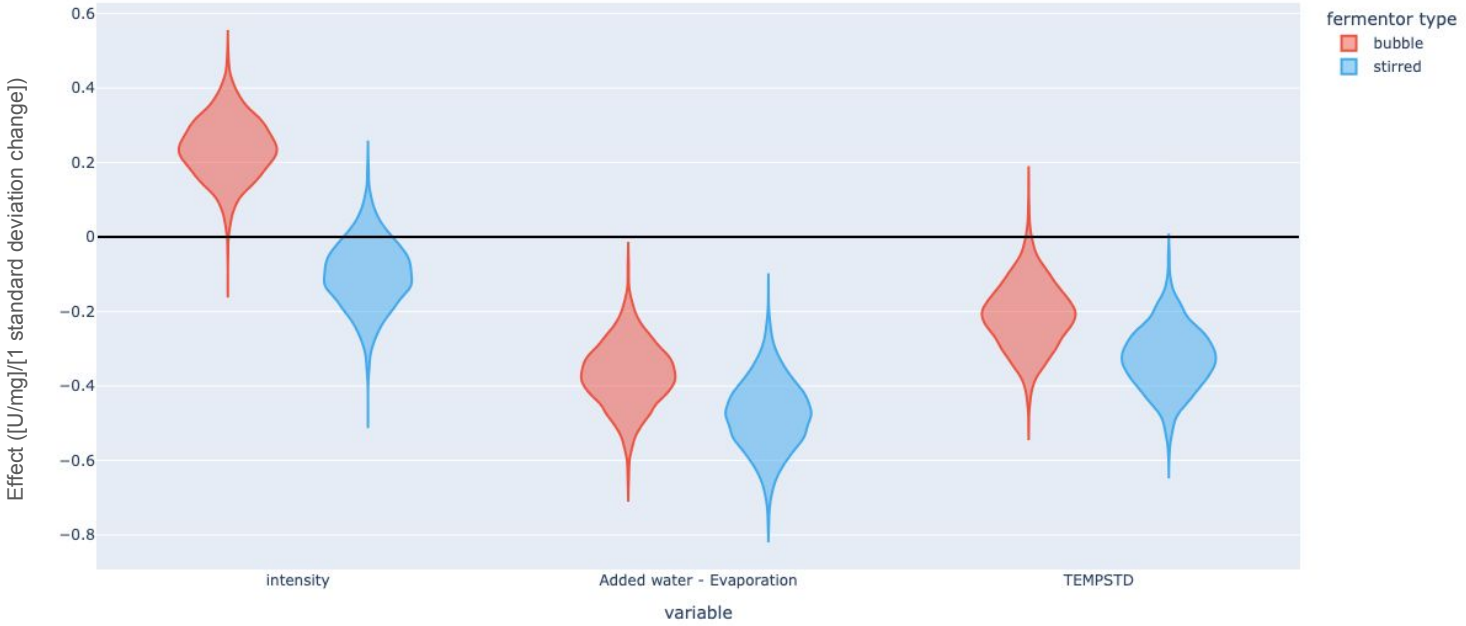
- Total water added is dependent on the air water content
- Biomass is affected by:
  - total water added
  - air water content
  - intensity
- Larger biomass → worse yield
- Larger biomass → more CO2

yield ~ intensity
+ net water
+ temp. std

DAG - a more realistic causal structure

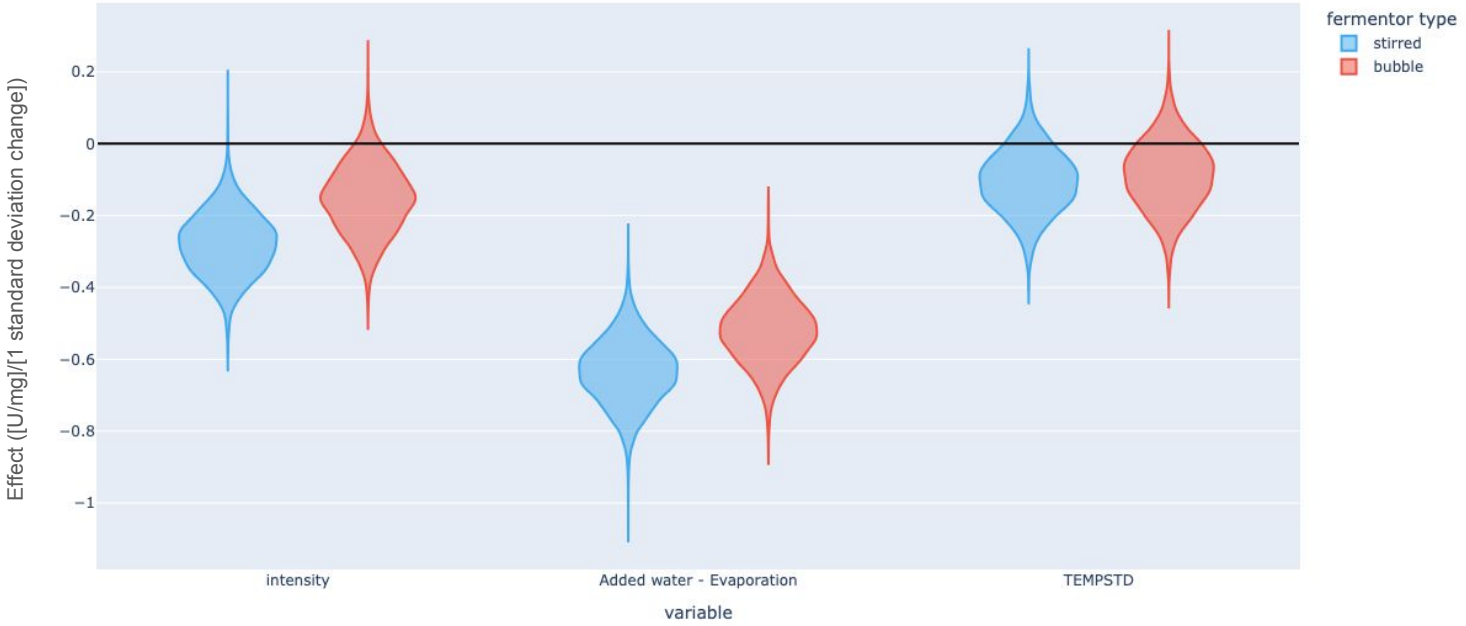


Yield coefficients for 2023 data



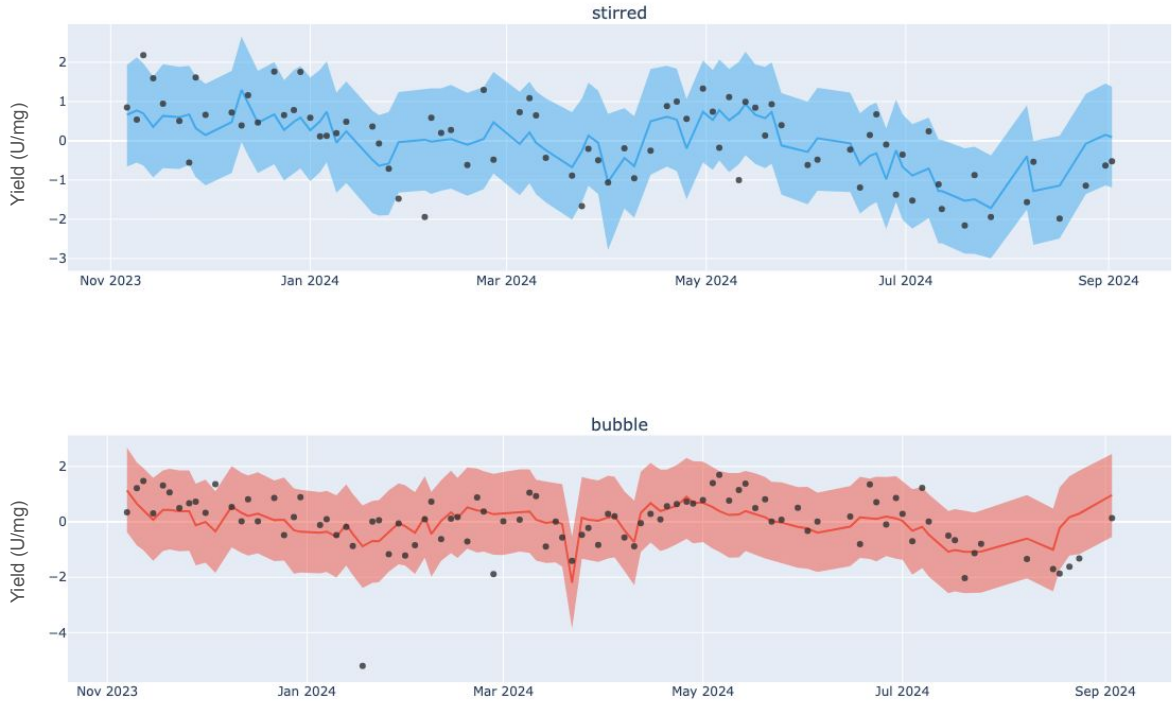
Multiple linear regression for 2024 shows significant negative effects of both temperature st. dev. and net water.

Yield coefficients for 2024 data



Multiple linear regression for 2024 shows weak negative effect of temperature, but significant negative effect of net water.

Predictive check for the yield model for 2024



*shaded regions present 95% credible intervals

The predictive check of the model based on the (i) intensity, (ii) net water, and (iii) temp. st. dev. captures the trends in yield well.

The solution and suggestions

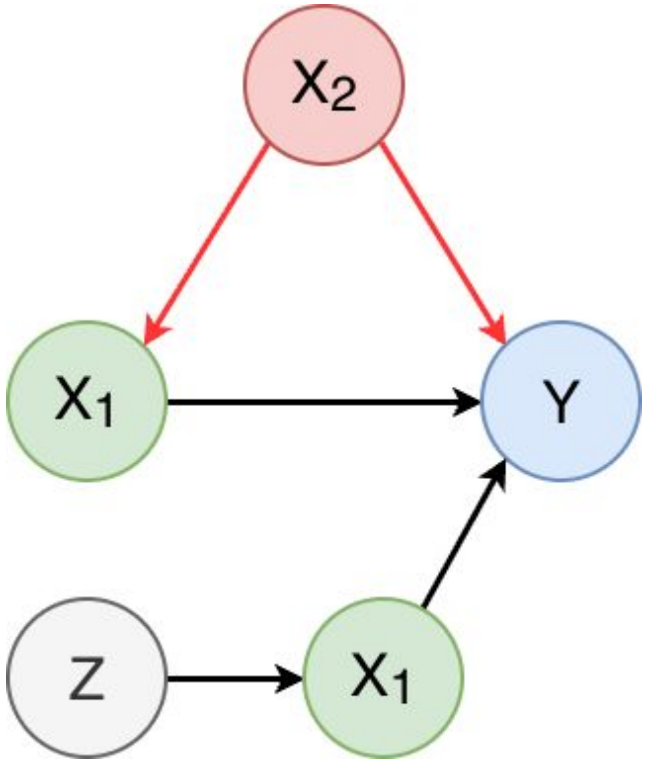
Required Water Calculator			
Inputs			
air water content	5.00	g/kg	*The most recent value of the air water content (e.g. average during previous batch/last few hours).*
desired water level	0.00	ton	*The desired total water level at the EOF (added water - evaporation). Suggested: 0 ton.*
fermentation length	180.00	h	*Estimated fermentation length in hours.*
fermentor type	bubble		*Choose either stirred of bubble.*
Outputs			
estimated evaporation	22.32	ton	*The estimated total evaporation over the course of fermentation.*
total water to add	22.32	ton	*The total amout of water added required to meet the inserted desired water level.*
required water flow rate	124.00	liters/h	*The required water flow rate to add every hour of fermentation to meet the inserted desired water level.*

\+ 🔬 controlled experiment design to actually verify assumed causality

Part II:Theory

What is a DAG?

- A Directed Acyclic Graph (DAG) is a graphical representation of assumed causal relationships.
- Nodes represent variables.
- Directed edges represent causal influence.
- Acyclic means no feedback loops: causes precede effects.



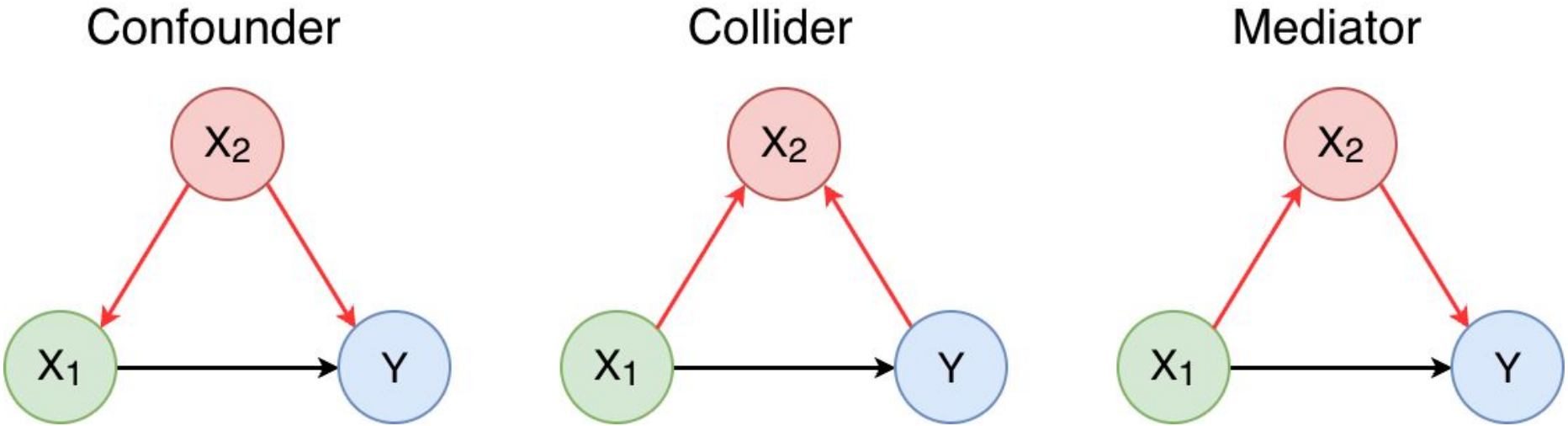
Why use a DAG?

- **Make causal assumptions explicit** before fitting regression or GLM models.
- Distinguish association from causation in **observational data**.
- **Guide variable selection**: what to adjust for and what not to adjust for.
- Clarify which parameters have a causal interpretation and which do not.
- Prevent common modeling mistakes that reduce interpretability or **induce bias**.

Key Points

- Regression coefficients admit a causal interpretation conditional on the **assumed DAG**.
- Misspecified causal structure **implies biased or meaningless estimates**.
- Adjusting for the wrong variables can introduce bias rather than remove it.
- At best it explains variance, at worst it produces wrong conclusions.
- DAGs are **not learned from the data**, they are primarily **driven by domain knowledge**, theory, and mechanistic understanding.

Common causal structures



Simple example: shark attacks









Data-generating model

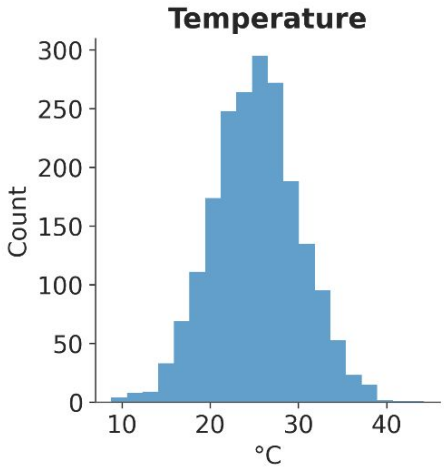
$$T_i \sim \mathcal{N}(25,\ 25)$$

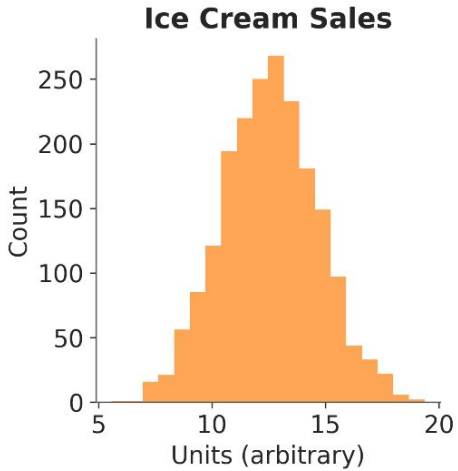
$$\mathrm{I}_i \mid T_i = 5.0 + 0.3\,T_i + \varepsilon_i \qquad \varepsilon_i \sim \mathcal{N}(0,\ 2.25)$$

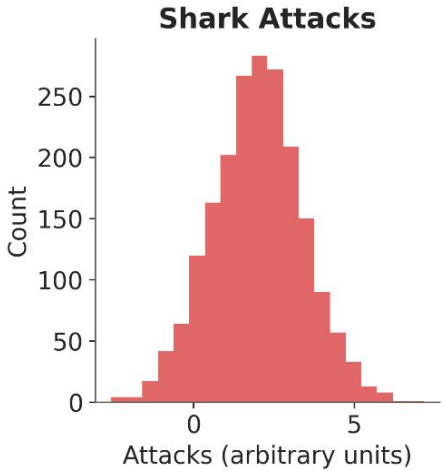
$$\mathrm{S}_i \mid T_i = -3.0 + 0.2\,T_i + \varepsilon_i \qquad \varepsilon_i \sim \mathcal{N}(0,\ 1.0)$$

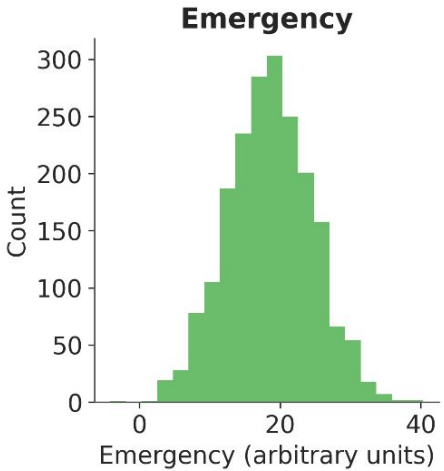
$$\mathrm{E}_i \mid I_i, S_i = 1.0\,I_i + 3.0\,S_i + \varepsilon_i \qquad \varepsilon_i \sim \mathcal{N}(0,\ 4.0)$$

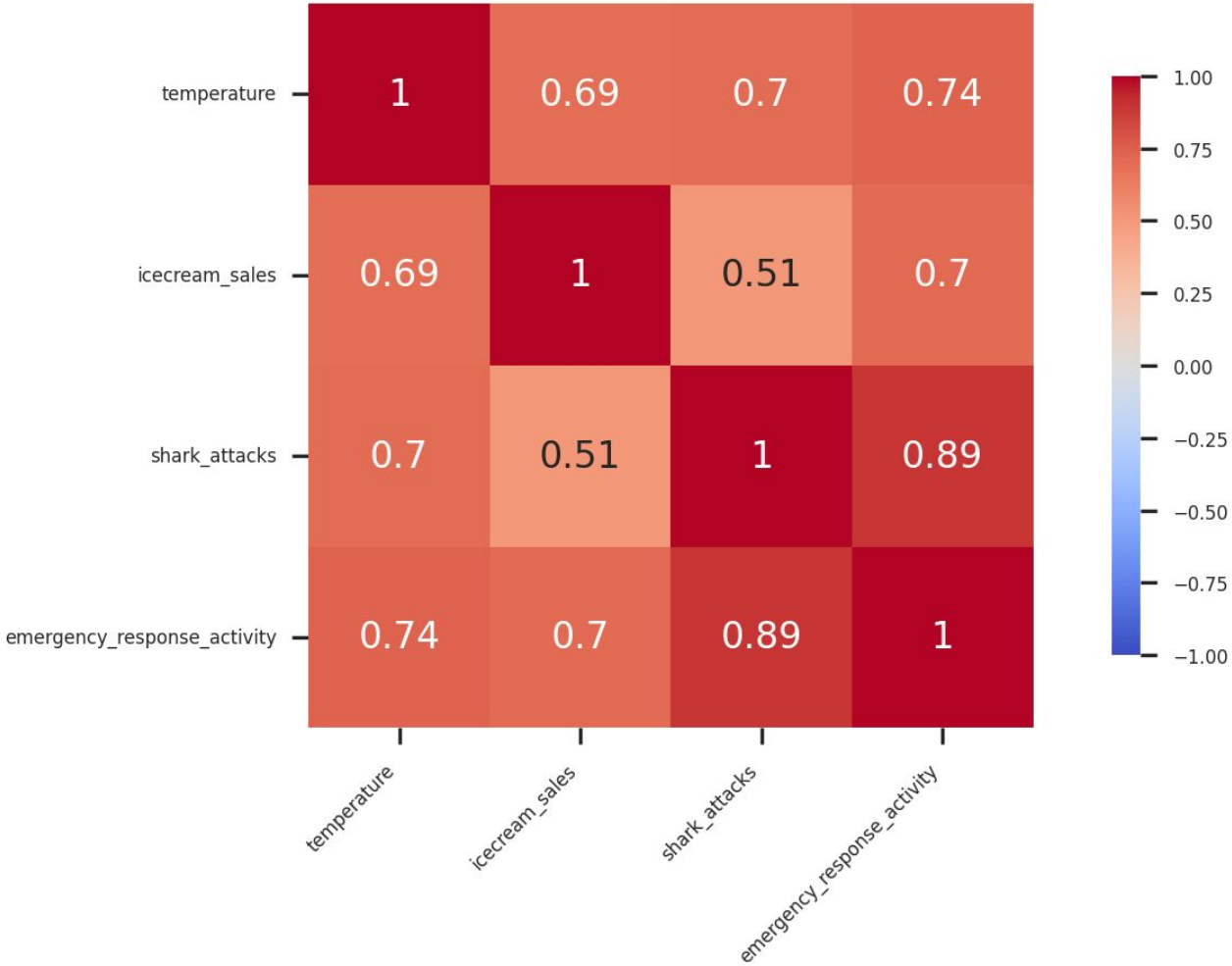
I … 🍦
S … 🦈
E … 🚑
T … 🌡️



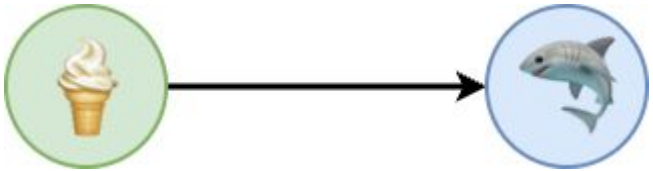






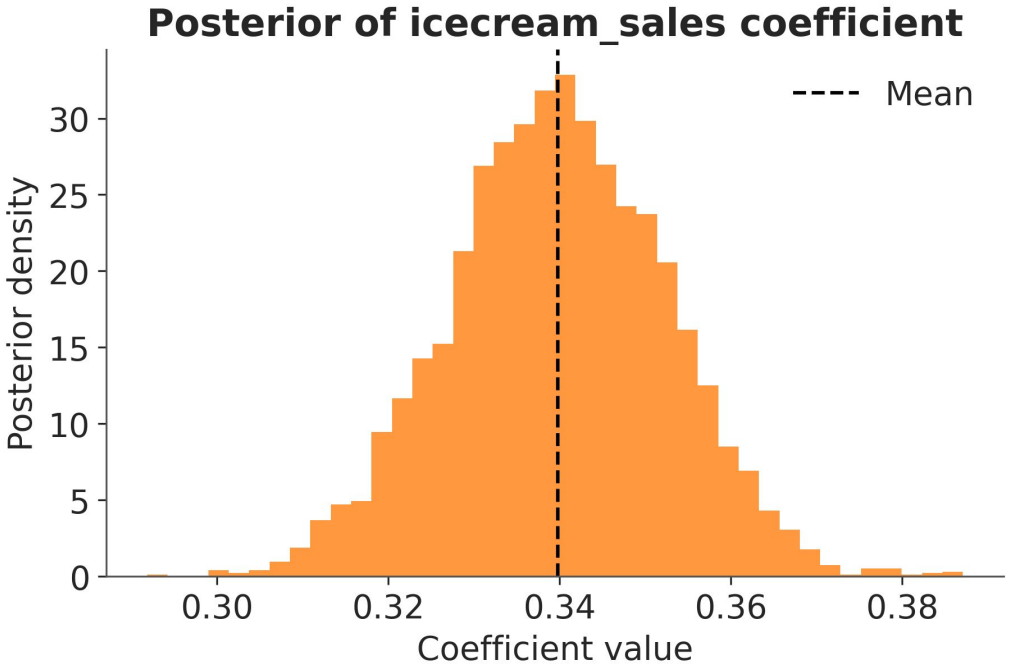

temperature
icecream_sales
shark_attacks
emergency_response_activity
1
0.69
0.7
0.74
0.69
1
0.51
0.7
0.7
0.51
1
0.89
0.74
0.7
0.89
1
temperature
icecream_sales
shark_attacks
emergency_response_activity
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

Model I: a simple and naive model

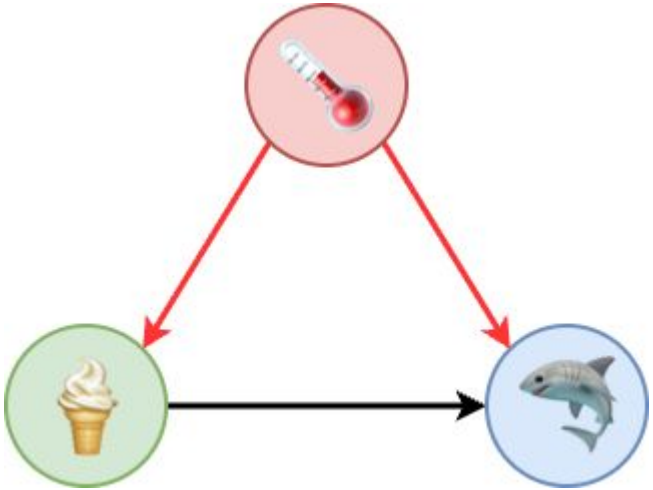


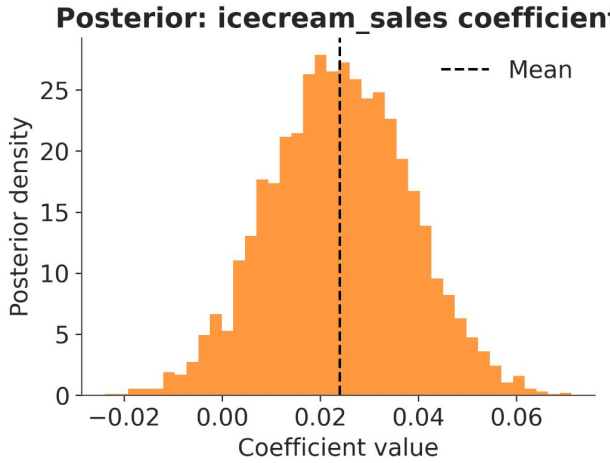
$$S_i \mid I_i \sim \mathcal{N}\left(\mu_i,\ \sigma_S^2\right)$$

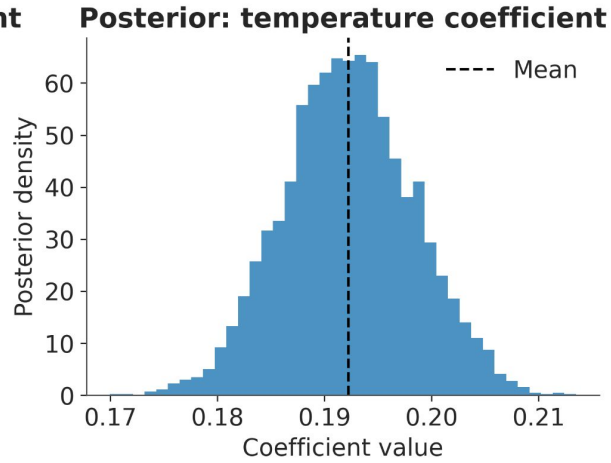
$$\mu_i = \alpha + \beta\, I_i$$



Model II: the model from common sense (i.e. experts)



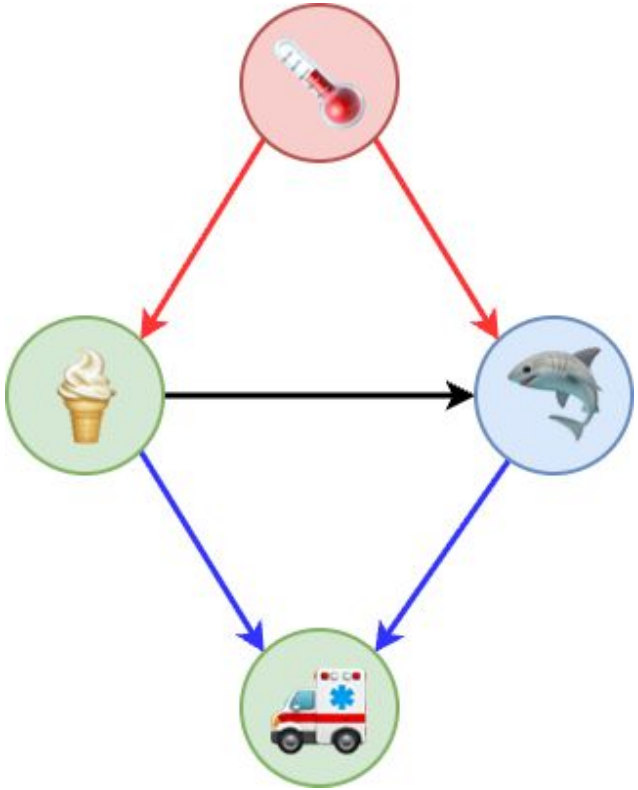




$$S_i \mid I_i, T_i \sim \mathcal{N}\left(\mu_i,\ \sigma_S^2\right)$$

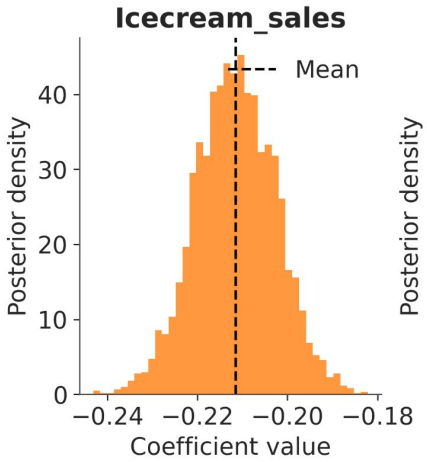
$$\mu_i = \alpha + \beta_I\, I_i + \beta_T\, T_i$$

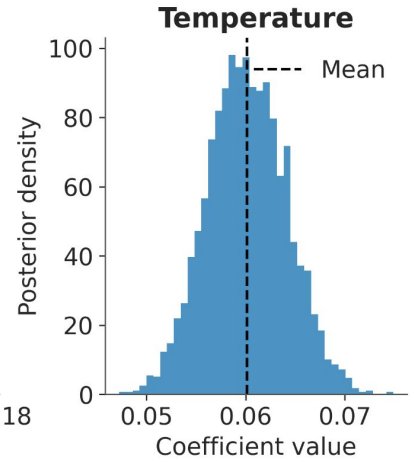
Model III: throwing everything we have at the model

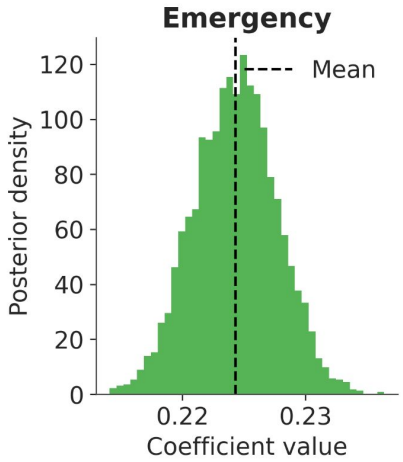


$$S_i \mid I_i, T_i, E_i \sim \mathcal{N}(\mu_i,\ \sigma_S^2)$$

$$\mu_i = \alpha + \beta_I\, I_i + \beta_T\, T_i + \beta_E\, E_i$$







Questions?